

CURRICULUM VITAE

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RESEARCH INTERESTS:

My primary area of research is the theory of turbulence in fluids and plasmas, with a special focus on the use of machine learning and reinforcement learning tools to solve difficult problems and, where possible, help elucidate the underlying physics. I am increasingly interested in using such data-driven tools to study complex animal behavior, and in particular how it emerges from physical properties and constraints of the environment. Among other topics, I have published work on source tracking in turbulent flows via Bayesian methods, transport model reduction using deep supervised learning, and the analytical theory of turbulent transport and self-organization in magnetized plasmas.

POSITIONS HELD:

MaLGa and DICCA, University of Genoa

Mar. 2025 – present

Postdoctoral fellow

Dept. Physics, University of Rome, “Tor Vergata”

Mar. 2021 – Mar. 2025

Postdoctoral fellow

Dept. Physics, University of California, San Diego

Apr. 2018 – Feb. 2021

Graduate research assistant

XCP Division, Los Alamos National Laboratory

Jan. – Aug. 2015

Post-baccalaureate intern

EDUCATION:

University of California, San Diego

Sep. 2015 – Feb. 2021

Ph.D. Physics (*with computational specialization*); GPA 3.810/4

Thesis Advisor: Patrick H. Diamond

Thesis Title: “Topics in mesoscopic turbulent transport.”

University of Minnesota, Twin Cities

Sep. 2010 – Jun. 2014

B.S. Physics and Mathematics (*magna cum laude*); GPA 3.819/4

Honors Thesis Advisor: Willard Miller, Jr. (deceased)

Thesis Title: “Structure equations for 2D 2nd order superintegrable systems.”

PUBLICATIONS:

[19] M. Carbone, L. Piro, **R. A. Heinonen**, L. Biferale, M. Cencini, and A. Celani. “Olfactory pursuit: catching a moving target via odor cues in complex flows.” (submitted to *Proc. Nat. Acad. Sci.*)

[18] **R. A. Heinonen**. “Asymptotics of solutions to the linear search problem.” arXiv: 2602.20991 (submitted to *Oper. Res.*)

[17] L. Piro, **R. A. Heinonen**, M. Carbone, M. Cencini, and L. Biferale. “Policy heterogeneity improves collective olfactory search in 3-D turbulence.” *Phys. Rev. E* **113**, 044401 (2026).

[16] L. Biferale, F. Bonaccorso, N. Cocciaglia, **R. A. Heinonen**, and L. Piro. “TURB-Smoke. A database of Lagrangian pollutants emitted from point sources in turbulent flows with a mean wind.” *Sci. Data* (accepted). arXiv: 2507.22749

[15] R. Varennes, Z. Qu, Y. W. Cho, C. Wan, K. Li, **R. A. Heinonen**, and V. Grandgirard. “A robust data-driven approach for modeling turbulent transport.” *Nucl. Fus.* **65**, 076027 (2025).

[14] **R. A. Heinonen**, L. Biferale, A. Celani, and M. Vergassola. “Exploring Bayesian olfactory search in

realistic turbulent flows.” *Phys. Rev. Fluids* **10**, 064614 (2025).

[13] L. Piro, **R. A. Heinonen**, M. Cencini, and L. Biferale. “Many wrong models approach to localize an odor source in turbulence with static sensors.” *J. Turbul.*, 1–21 (2025).

[12] **R. A. Heinonen**, L. Biferale, A. Celani, and M. Vergassola. “Optimal trajectories for Bayesian olfactory search in turbulent flows: The low-information limit and beyond.” *Phys. Rev. Fluids* **10**, 044601 (2025).

[11] **R. A. Heinonen**, L. Biferale, A. Celani, and M. Vergassola. “Optimal policies for Bayesian olfactory search in turbulent flows.” *Phys. Rev. E* **107**, 055105 (2023).

[10] A. Loisy and **R. A. Heinonen**. “Deep reinforcement learning for the olfactory search POMDP: a quantitative benchmark.” *Euro. Phys. J. E* **46**, 17 (2023).

[9] **R. A. Heinonen**, P. H. Diamond, M. F. D. Katz, and G. E. Ronimo. “Generation of momentum transport in weakly turbulent β -plane magnetohydrodynamics.” *Phys. Rev. E* **107**, 025202 (2023).

[8] O. Heinonen, **R. A. Heinonen**, and H. Park. “Magnetic ground states of a model for $M\text{Nb}_3\text{S}_6$ ($M = \text{Co}, \text{Fe}, \text{Ni}$).” *Phys. Rev. Mat.* **6**, 024405 (2022).

[7] **R. A. Heinonen** and P. H. Diamond. “Learning how structures form in drift-wave turbulence.” *Plasma Phys. Control. Fusion* **62**, 105017 (2020).

[6] **R. A. Heinonen** and P. H. Diamond. “Turbulence model reduction by deep learning.” *Phys. Rev. E* **101**, 061201(R) (2020).

[5] **R. A. Heinonen**, D. Saumon, J. Daligault, C. E. Starrett, S. D. Baalrud, and G. Fontaine. “Diffusion coefficients in the envelopes of white dwarfs.” *Astrophys. J.* **896.1**, 2 (2020).

[4] **R. A. Heinonen** and P. H. Diamond. “A closer look at turbulence spreading: How bistability admits intermittent, propagating turbulence fronts.” *Phys. Plasmas* **27**, 032303 (2020).

[3] **R. A. Heinonen** and P. H. Diamond. “Subcritical turbulence spreading and avalanche birth.” *Phys. Plasmas* **26**, 030701 (2019).

[2] N. M. Gill, **R. A. Heinonen**, C. E. Starrett, and D. Saumon. “Ion-ion dynamic structure factor of warm dense mixtures.” *Phys. Rev. E* **91**, 063109 (2015).

[1] **R. A. Heinonen**, E. G. Kalnins, W. Miller, Jr., and E. Subag. “Structure relations and Darboux contractions for 2D 2nd order superintegrable systems.” *SIGMA* **11**, 043 (2015).

CONFERENCE TALKS:

[17] EFDC (2025). “(Mostly) Bayesian olfactory search in turbulent flows: correlation effects.”

[16] EFDC (2024). “Bayesian olfactory search in realistic turbulent flows.”

[15] APS-DFD (2023). “Optimal Bayesian olfactory search in a realistic turbulent flow.”

[14] ETC (2023). “Optimal policies for Bayesian olfactory search in a turbulent flow.”

[13] APS-DFD, (2022). “Data-driven Bayesian olfactory search in a turbulent flow.”

[12] EFMC (2022). “Optimal policies for Bayesian olfactory search in a turbulent environment.”

[11] Festival de Théorie (2022). “Optimal policies for olfactory search in a turbulent environment using POMDP.”

[10] APS-DFD, (2021). “Search strategies in a turbulent flow using a POMDP framework.”

[9] AAPPs-DPP, invited (2021). “On the role of cross-helicity in β -plane MHD turbulence.”

[8] Euromech Colloquium 614, “Machine learning methods for prediction and control of turbulent flows” (2021). “Learning how structures form in drift-wave turbulence.”

[7] APS-DPP (2020). “On the role of cross-helicity in β -plane MHD turbulence.”

[6] APS-DPP (2020). “Turbulence model reduction by deep learning.”

- [5] AAPPS-DPP, invited (2020). “Turbulence model reduction by deep learning.”
- [4] Transport Task Force (2020). “Turbulence model reduction by deep learning” (cancelled due to COVID-19).
- [3] APS-DPP, invited (2019). “Subcritical turbulence spreading and avalanche birth.”
- [2] APS-DPP, (2019). “Learning a model for mean-field turbulence dynamics.”
- [1] Festival de Théorie (2019). “Subcritical turbulence spreading and avalanche birth.”

CONFERENCE POSTERS:

- [6] Complex Flows and Complex Fluids, Satellite StatPhys29 (2025). “Plume-loss trajectories for olfactory search in turbulent flows.”
- [5] Transport Task Force (2019). “Learning mean field dynamics of drift-wave turbulence.”
- [4] APS-DPP (2018). “A closer look at turbulence spreading: how bistability admits intermittent, propagating turbulence pulses.”
- [3] Chengdu Theory Festival (2018). “A closer look at turbulence spreading: how bistability admits intermittent, propagating turbulence pulses.”
- [2] Transport Task Force (2018). “A closer look at turbulence spreading: how bistability admits intermittent, propagating turbulence pulses.”
- [1] LANL 15th Annual Student Symposium (2015). “Coefficients of diffusion in dense stellar plasmas.”

SEMINARS:

- [3] *Physical Review* Journal Club (2025). “Optimal olfactory search in turbulent flows.”
- [2] Invited seminar, University of California, Santa Barbara (2023). “Optimal policies for Bayesian olfactory search in a turbulent flow.”
- [1] Young Seminars SIFS, Italian Society of Statistical Physics (2023). “Optimal policies for Bayesian olfactory search.”

SELECTED AWARDS/HONORS:

- Invited Talk, *Physical Review* Journal Club (“Optimal trajectories for Bayesian olfactory search in turbulent flows: The low-information limit and beyond”), 2025
- *Phys. Rev. Fluids* Editor’s Suggestion (“Optimal trajectories for Bayesian olfactory search in turbulent flows: The low-information limit and beyond”), 2025
- Invited Talk, AAPPS-DPP (“On the role of cross-helicity in β -plane MHD turbulence”), 2021
- Invited Talk, AAPPS-DPP (“Turbulence model reduction by deep learning”), 2020
- Invited Talk, APS-DPP (“Subcritical turbulence spreading and avalanche birth”), 2019
- Fellow, Aix-en-Provence Festival de Théorie, 2019
- *Physics of Plasmas* Featured Article and Scilight (“Subcritical turbulence spreading and avalanche birth”), 2019
- Fellow, LANL Computational Physics Student Summer Workshop, 2014
- Phi Beta Kappa, 2014
- Hans H. Dalaker Scholarship in Mathematics, 2012 & 2013
- Alfred O. C. Nier Scholarship in Physics, 2012

PEER REVIEW:

Referee for *Environmental Data Science*, *Physical Review Fluids*, *Journal of Plasma Physics*, *Archives of Computational Methods in Engineering*, *Scientific Reports*, *Physical Review E*, *Nuclear Fusion*, *Physics of Plasmas*, *Machine Learning: Science and Technology*, *Nonlinear Dynamics*, *EPL*, and *Communications Physics*.

ERDŐS NUMBER: 3

Via W. Miller, Jr., and L. A. Rubel.